

CIS 5030 Algorithms for Big Data

Summer 2026

Instructors

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Course Description

This course focuses on algorithmic techniques needed for processing large amounts of data. A traditional course in algorithm design teaches foundational principles (divide-and-conquer, dynamic programming, graph algorithms) that are primarily suited for solving computational problems in polynomial time. What happens when traditional techniques meet modern scales, and “polynomial”-time is too slow? This course will set the mathematical and algorithmic foundations for processing large amounts of data. The key will be to incorporate randomness and approximation and to develop the algorithmic techniques needed to keep up with ever-increasing scales.

The successful student will leave this class with a theoretical computer science perspective on problems and models for massive data. They will gain knowledge of various algorithmic techniques and reasoning tools to design algorithms using randomness and approximation. They will furthermore learn the fundamental limits of these techniques. This aligns with the Computer Science program objective to produce graduates with the ability to successfully apply analytical and problem-solving skills, and to reason about computational approaches for algorithmic problems with modern scales.

Course Learning Objectives

- Identify, formulate, and solve complex algorithmic problems in resource-constrained environments from a theoretical computer science perspective.

- Design and analyze known and new algorithms using randomness and approximation.
- Communicate effectively in a variety of professional contexts.
- Recognize resource-constrained environments and make informed judgments on algorithmic principles and performance of algorithms.
- Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.

Course Prerequisites

- CIT 5920 required
- Suggested: An undergraduate algorithms course (like CIS 1210 or equivalent course), or comfort with models of computation, asymptotic complexity, and basic probability.

Course Textbook

No required texts. Materials will be distributed throughout the course.

Grading & Assessment

You must attempt all graded assignments to pass the course. If you have any questions or concerns about grading or progress in the course, please reach out to your instructor.

This course will use a variety of assessments to determine whether learners understand and can apply the key concepts and skills that the course teaches. This includes:

Type	%	Description
Quizzes	10%	You'll complete twelve quizzes in this course, one following each of Modules 1–13 (none for Module 12). Quizzes are automatically graded and count toward your final grade, and you'll have unlimited attempts on each one.
Homework Assignments	50%	There are a total of six graded homework assignments, with one due every two modules. All homework must be submitted through Gradescope, where TAs will grade and provide feedback.

Exam 1	20%	You'll complete two proctored exams, one at the end of Module 7 and another in Module 14. Exams are administered using Honorlock, a live online AI-based proctoring system.
Exam 2	20%	

Please read the instructions for each assignment very carefully to make sure you know where to submit to receive credit!

Grade Ranges

The cutoffs for letter grades will be decided based on the class performance on various grading components for the course (quizzes, homeworks, and exams).

Late Policy

The instruction staff is committed to your success and understands how challenging it can be to learn online while balancing other commitments. Despite students' best intentions, sometimes life gets in the way, and a little extra time to complete an assignment may be necessary.

If you need extra time on an assignment, you can obtain an extension for extenuating circumstances. If an extension is not approved, an assignment that is turned in late will receive a **late penalty of 10% of the total possible points per day, up to 3 days**. After the 3rd day, no credit will be given.

For extensions, please request them via the **Extension Request Form** on the Canvas course homepage. These extension requests must be submitted at least 24 hours before the assignment deadline. If your request is granted, you will see updated deadlines reflected in Canvas and Gradescope, and you will receive an email confirming the status of your extension request. You must fill out this form for each assignment you wish to have an extension for.

Regrade Requests

Regrade requests are handled on a case-by-case basis and must be made using the **Regrade Request function in Gradescope**. You have up to 5 days after grades are released to submit a request.

Regrade requests may take up to a week to process at the discretion of the faculty. When submitting a regrade request, please explain (in detail) why you feel the grading

is incorrect. Keep in mind that a regrade can result in a higher score, a lower score, or no change.

Extra Credit

There are no extra credit opportunities planned. If this changes, any updates will be posted on Ed Discussion.

Other Course Activities

The following activities are not mandatory, but will greatly support your success on the graded assignments.

Discussion Forum

Discussion forums (on Ed Discussion) are designed to give you optional extra practice with the material and to see examples of how your classmates are thinking and working. In addition, you can use the discussion forums to ask questions about upcoming assignments and logistical concerns. Course staff will make their best effort to reply within 24 hours to these posts. **Effectively monitoring Ed Discussion for staff announcements, assignment updates, and adjustments to the course schedule is essential for being successful in this course.**

Ed Discussion allows for both public and private posting of questions, comments, and concerns. Private posts should be used anytime you need to discuss sensitive information such as grades, personal issues, or solutions to assignments or exams. Due to the volume of Ed Discussion posts received on a daily basis, our course staff prioritizes the responses to public posts as well as private posts containing sensitive information. It is highly encouraged that you create a public post when discussing information that could benefit a wider audience, such as assignments and course logistics, or content-related questions. If you wish to create a public post without having your name attached to it you can create a public post anonymously. Please be aware that, course staff reserves the right to flip private posts to public and change your identity to anonymous if the post does not contain sensitive information and our course staff believes the content of your post could be beneficial to other students.

Our program's **AI chatbot, Jean**, is a part of every course's Ed Discussion. For all up-to-date guidance on Jean, please see the [Student Guidelines](#) found within the Canvas course.

Recitation

Recitations are biweekly live sessions with your TAs designed around some kind of problem/activity that is supposed to take about an hour to solve. TAs will answer questions as you work through the problem or answer questions that you have submitted in advance of the session. If you are not able to attend either of the scheduled recitation times, you can review a recording of the session, posted the day after.

Additional Segments

The professor may add additional optional segments to support the class as needed.

Creating a Welcoming Environment

All members of the course community – the instructor, TAs, and students – are expected to work together to create a supportive, inclusive environment that welcomes all students, regardless of their race, color, sex, sexual orientation, religion, creed, national origin (including shared ancestry or ethnic characteristics), citizenship status, age, disability, veteran status or any other class protected under applicable federal, state, or local law. **All participants in this course deserve to and should expect to be treated with respect by other members of the community.**

Discussion boards, messaging channels, recitations, office hours, and group working time should be spaces where everyone feels welcome and safe. In order to facilitate a welcoming environment, students of this course are expected to:

- Exercise consideration and respect in their speech and actions
- Attempt collaboration and consideration, including listening to opposing perspectives and authentically and respectfully raising concerns, before conflict
- Refrain from demeaning, discriminatory, or harassing behavior and speech

All members of the course community are expected to be familiar with and abide by the University's guidelines on general conduct and sexual harassment:

- University Code of Conduct:
<https://catalog.upenn.edu/pennbook/code-of-student-conduct/>
- University Sexual Misconduct Policy:
<https://catalog.upenn.edu/pennbook/sexual-misconduct-resource-offices-complaint-procedures/>

Students should also be familiar with other University guidelines regarding personal conduct:

- Conduct & Personal Responsibility guidelines in Pennbook:
<https://catalog.upenn.edu/pennbook/#policiesbytopictext>
- University Principles of Responsible Conduct:
http://www.upenn.edu/audit/oacp_principles.htm

If you are a victim of, witness, or are otherwise affected by unacceptable behavior:

- In cases of sexual harassment or assault, please consult DPS Special Services (<https://www.publicsafety.upenn.edu/about/special-services/sensitive-crimes/> at 215-573-3333; this is a confidential resource.
- To report other bias incidents, please complete the Bias Incident Reporting Form: <https://belonging.upenn.edu/diversity-at-penn/bias-motivated-incident-report>
- For other violations of the code of student conduct, the Center for Community Standards and Accountability has an incident reporting form at <https://csa.upenn.edu/community-standards/refer-caserequest-consultation>

If you are unsure which office to contact, please contact the instructor or any Penn Engineering Online Learning staff member.

Getting and Giving Help

TA and Faculty Support

TAs will hold office hours weekly, when they will offer 15-minute time slots on a sign-up basis. Your professor will be available for a limited number of private meetings per week, depending on the needs of the class.

Collaboration Guidelines

You can collaborate on each homework with up to 2 other students (i.e. groups can be of size 3). This collaboration allows you to discuss ways to solve a problem. Collaborators may submit anything from the exact same document, to completely different documents, but in case of identical submissions, all collaborators **must make a single assignment submission**. In case of identical submissions, be sure to add all collaborators on Gradescope when submitting. While identical homework submissions are allowed for collaborating students, it is strongly recommended that collaborations are used only to discuss ideas and that each student writes homework solutions by themselves so as to learn the course material better. This is an important skill to develop and will be helpful in writing solutions on your own during exams. Students may change collaborators for each assignment if they wish.

Additionally, discussion forums and recitations *are* collaborative—please take advantage of those times to work with your colleagues. For general communication with your colleagues, use your Slack channels or Slack direct messages.

Forming study groups to understand the material is also a good idea, as long as groups stay on the conceptual level and are not collaborating on the graded assignments directly.

Note: When in doubt always ask the instructors or TA first to avoid any potential collaboration that can be construed as academic dishonesty.

Do not cheat. **Please note that searching for solutions online is the same as cheating. Posting solutions online is also considered cheating.** If you are caught posting solutions or code to a publicly accessible location (like StackOverflow or GitHub), it will be considered cheating. If you do use GitHub (or similar cloud-based code management system) to set up a remote code repository, **YOU ARE REQUIRED TO KEEP THAT REPOSITORY PRIVATE.** Accessing solutions from previous iterations of this course is considered cheating, as is the use of generative AI tools to complete course assignments.

You can further read Penn's [Code of Academic Integrity](#) page on this subject matter, as well as the SEAS Graduate Student guidelines on the code of ethics.

Guidelines for the Use of Generative AI in this Course

We recognize the increasing prevalence and power of Artificial Intelligence (AI) tools, and encourage their responsible and ethical use to enhance your learning experience. However, it's essential to develop a strong understanding of fundamental processes before relying on AI. To that end:

Examples of acceptable uses of AI

- Comprehension and Expansion: You are encouraged to use AI to clarify and expand your understanding of course materials.
- Research and Information Gathering: AI can supplement your research and information gathering, aiding your exploration of complex concepts.

Examples of unacceptable uses of AI

- Assignment Completion: Avoid using AI for generating code or text for assignments, peer reviews, or content summaries.
- Assessment Assistance: Do not seek AI help for quizzes, exams, or assessments.

- **Academic Integrity:** Do not present AI-generated content without citation and as your own work.

Engaging in unacceptable use of AI tools will result in academic consequences, which may include grade penalties, academic warnings, or other actions as determined by your instructor and the university's academic integrity policies.

Note that these guidelines may differ from those in other courses. If you have questions or concerns, don't hesitate to reach out. Ultimately, AI can be a valuable educational tool when used responsibly and aligned with our course policies.

Plagiarism Policy

Students found plagiarizing on any assignment will automatically receive a zero on that assignment. The first offense of plagiarism on an assignment will be handled by the instructor. Only in extenuating circumstances, or if the plagiarism was committed on an exam, will a first offense be turned over to the University of Pennsylvania Center for Community Standards and Accountability.

A second offense, or, exam plagiarism, will be turned over immediately to the University of Pennsylvania Center for Community Standards and Accountability.

A report being submitted to the University of Pennsylvania Center for Community Standards and Accountability will result in a failure of the course (regardless of current course average), and a potential permanent notation on your academic record that will follow you to all future academic institutions and possibly future employers. If you are unfamiliar with what constitutes plagiarism at Penn, visit Penn's [Code of Academic Integrity](#).

Recording Notice

Public office hours, recitations, and other live events will be recorded, used, and may be made available to class participants during the current semester as well as students who take the class in future semesters.

Private office hours will also be offered and are not recorded. Students who do not wish to attend the publicly recorded office hours may attend the private office hours.

Access to Materials and Content Before and After Graduation

If you would like to retain copies of your submitted assignments, you must download them from Gradescope, Canvas, Codio, and any other platforms that you submit to during the semester in which you are taking that course.

Access to course materials and your submissions is not guaranteed after the completion of a course. Therefore, we recommend that students download any assignments or materials they would like to keep before a course concludes.

Summer 2026 Course Schedule and Important Dates

Dates are subject to change. Please check Ed Discussion for announcements regarding schedule changes. Note: Weeks run Monday through Sunday.

Assignment Submissions:

- **Quizzes:** Quizzes are due at the end of the term, though it's recommended to complete them by the Sunday of the week they cover.
- **Homework:** All homework assignments will be made available two weeks prior to their due dates. Assignments are due on alternate Tuesdays (13 days after they open).
- **Exams:** You will have 3 days to submit your exams from the time they become available.

Monday, 5/4 - Sunday, 5/10	Module 1	Topic(s) Probabilistic Preliminaries	Assignment(s) Quiz 1 HW1 released May 6, 2026
Monday, 5/11 - Sunday, 5/17	Module 2	Topic(s) Estimation via Sampling	Assignment(s) Quiz 2 HW1 due Tuesday May 19, 2026
Monday 5/18 - Sunday 5/24	Module 3	Topic(s) Hashing and Bloom Filters	Assignment(s) Quiz 3 HW2 released May 20, 2026
Tuesday 5/26 - Sunday 5/31 Memorial Day 5/25 - No Class	Module 4	Topic(s) Compressing Vectors with Sketching	Assignment(s) Quiz 4 HW2 due Tuesday June 9, 2026
Monday 6/1 - Sunday 6/7	Module 5	Topic(s) Hashing-Based Estimation	Assignment(s) Quiz 5 HW3 released June 10, 2026

Monday, 6/8 - Sunday, 6/14	Module 6	Topic(s) Locality-Sensitive Hashing	Assignment(s) Quiz 6 HW3 due Tuesday June 23, 2026
Monday, 6/15 - Sunday, 6/21 Juneteenth 6/19 - No Class	Module 7	Topic(s) Property Testing	Assignment(s) Quiz 7 Exam 1: June 19 – June 21, 2026 HW 4 released: June 24, 2026
Monday, 6/22 - Sunday, 6/28	Module 8	Topic(s) Linear Algebra Preliminaries	Assignment(s) Quiz 8 HW 4 due: July 7, 2026
Monday, 6/29 - Sunday, 7/5 Independence Day 7/3 - No Class	Module 9	Topic(s) Linear Regression	Assignment(s) Quiz 9 HW 5 released: July 8, 2026
Monday, 7/6 - Sunday, 7/12	Module 10	Topic(s) The Power Method	Assignment(s) Quiz 10 HW 5 due: July 21, 2026
Monday, 7/13 - Sunday, 7/19	Module 11	Topic(s) SVD and Application	Assignment(s) Quiz 11 HW6 released: July 22, 2026
Monday, 7/27 - Sunday, 8/2	Module 12	Topic(s) Linear programming	Assignment(s) HW6 due: August 4, 2026
Monday, 8/3 - Sunday, 8/9	Module 13	Topic(s) Dimension Reduction	Assignment(s) Quiz 13
Monday, 7/27 - Sunday, 8/2	Module 14	Topic(s) Closing Comments	Assignment(s) Exam 2: August 7 – August 9, 2026